

STCA-ViT: Spatio-Temporal Cross-Attention Vision Transformer for Video Action Recognition

Jiha Kim, Seunghyun Park and Hyunhee Park

Abstract—Video action recognition requires modeling complex spatial and temporal dependencies under strict efficiency constraints. This paper proposes STCA-ViT, a hybrid framework that integrates a VideoMAE ViT-Base backbone with bidirectional spatio-temporal cross-attention, Sobel-based boundary detection for motion gating, and YOLO-based person attention in parallel. The architecture incorporates residual contribution fusion with gradient reversal for feature diversity along with orthogonal regularization, a multi-head ensemble classifier, and an optional weakly-supervised temporal action localization (WTAL) head that recycles motion features to produce frame-level action scores without temporal annotations. Experimental results on standard benchmarks demonstrate the effectiveness of the proposed approach. STCA-ViT achieves a Top-1 accuracy of 90.46% on HMDB-51, surpassing VideoMAE V2 ViT-g (86.08%), while achieving 98.29% on UCF-101, 79.44% on Kinetics-400, and 65.01% on Something-Something-v2, using only 110M parameters and 305 GFLOPs. These results confirm that the proposed architecture achieves a favorable balance between accuracy and efficiency, making it suitable for real-time and embedded video systems. The proposed framework provides a promising foundation for extension toward multi-modal integration, long-range temporal context, and deployment in low-power or resource-constrained environments. The source code is publicly available at <https://github.com/yaki-toki/STCA-ViT>.

Index Terms—human action recognition, spatio-temporal modeling, vision transformer, attention mechanism, efficient video understanding

I. INTRODUCTION

VIDEO-BASED human action recognition plays a central role in a wide range of intelligent systems such as autonomous driving, surveillance, and smart cities. The ability to accurately and efficiently recognize the actions of pedestrians, cyclists, and vehicles is essential for safety-critical decision-making. Video understanding technology is also widely applied in fields such as sports analytics and human-computer interaction (HCI). As the scale and complexity of video data continue to grow, developing models that simultaneously satisfy high recognition accuracy and computational efficiency remains an important research challenge.

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This work was supported by the Ministry of Trade, Industry and Energy (MOTIE) and the Korea Evaluation Institute of Industrial Technology (KEIT), and the National Research Foundation of Korea (NRF) (Grant No.: 2022R1A2C2005705, AI-MAC Protocol for Intelligent Flying Base Stations Based on Distributed Machine Learning; RS-2024-00469138, Development of Vision Recognition Technology SDK Based on Domestic AI Semiconductors).

Existing methods primarily rely on 3D convolutional neural networks (3D CNNs) or transformer-based architectures to capture spatial semantics and temporal dynamics. While these models achieve competitive performance, they generally involve high computational costs and treat all spatio-temporal regions with equal importance. In real-world environments where background noise and action-irrelevant motion are present, this can degrade recognition performance. Furthermore, large-capacity architectures impose a burden on real-time inference and make deployment on embedded platforms difficult.

To address these issues, this work proposes STCA-ViT, a hybrid framework in which a VideoMAE ViT-Base backbone and four auxiliary modules process raw video input in parallel: (1) a bidirectional spatio-temporal cross-attention (STCA) module with learnable temporal tokens, (2) a Sobel-based Motion Gate Unit (MGU) incorporating multi-scale bidirectional motion and boundary detection, (3) a YOLO-based person attention module, and (4) an optional weakly-supervised temporal action localization (WTAL) head. The architecture also applies gradient reversal-based residual contribution fusion for module diversity, multi-head ensemble classification for robust prediction, and orthogonal loss regularization.

The main contributions of this work are as follows:

- **STCA module:** Bidirectional cross-attention between spatial and temporal tokens extracted via 3-stage Conv3d pathways with learnable temporal tokens and input-dependent gated fusion.
- **MGU:** Sobel-based gradient extraction with multi-scale bidirectional temporal differencing, boundary detection, and a learnable motion token for compact motion representation.
- **Residual Contribution Fusion:** Gradient reversal-based fusion combining backbone and three auxiliary features via sigmoid-gated residuals refined by 5-token self-attention, with a multi-head ensemble classifier for robust prediction.
- **Optional WTAL head:** Recycles MGU temporal features through a 5-component loss (MIL classification, pseudo-label supervision, sparsity, smoothness, background entropy) enabling frame-level action scoring without temporal annotations.

The remainder of this paper is organized as follows. Section II reviews related work on video transformers, motion modeling, and attention mechanisms. Section III describes the proposed STCA-ViT architecture in detail. Section IV presents experimental results and complexity analysis. Section V

TABLE I: Summary of notation used in the STCA-ViT network

Symbol	Definition	Dimension/Type	Description
$\mathbf{x}$	Input video tensor	$\mathbb{R}^{B \times 3 \times T \times H \times W}$	Input batch; B : batch size, T : number of frames, H, W : spatial size
D	Backbone embedding dimension	scalar	VideoMAE ViT-Base hidden size
D_m	Module embedding dimension	scalar (default D)	Dimension for STCA, MGU, PersonAttn
$\mathbf{b}_{back}$	Backbone features	$\mathbb{R}^{B \times D}$	Mean-pooled VideoMAE output
$\mathbf{S}_{tok}$	Spatial tokens	$\mathbb{R}^{B \times N_s \times D_m}$	3-stage Conv3d spatial extraction
$\mathbf{T}_{tok}$	Temporal tokens	$\mathbb{R}^{B \times N_t \times D_m}$	Conv3d temporal + learnable tokens
$\mathbf{f}_{stca}$	STCA module output	$\mathbb{R}^{B \times D_m}$	Gated spatio-temporal fusion
$\mathbf{M}_{enc}$	Motion encoder output	$\mathbb{R}^{B \times D_m \times T' \times H' \times W'}$	Encoded motion features
$\mathbf{B}_{mask}$	Boundary mask	$\mathbb{R}^{B \times 1 \times T' \times H' \times W'}$	Sigmoid boundary detection mask
$\mathbf{w}_{temp}$	Temporal attention	$\mathbb{R}^{B \times 1 \times T' \times 1 \times 1}$	Per-frame importance weights
$\mathbf{f}_{mgu}$	MGU output	$\mathbb{R}^{B \times D_m}$	Motion token from MGU module
$G(x, y)$	2D Gaussian map	scalar function	Attention at detected person center
$\mathbf{A}_{person}$	Person attention map	$\mathbb{R}^{B \times 1 \times T \times H \times W}$	YOLO-guided Gaussian attention
$\mathbf{f}_{person}$	Person feature vector	$\mathbb{R}^{B \times D_m}$	YOLO-guided attention-weighted pooling
α, β, γ	Contribution weights	scalar (sigmoid)	Learnable fusion weights for stca, mgu, person
$\mathbf{r}_i$	Module residual	$\mathbb{R}^{B \times D}$	FC(concat($\mathbf{f}_i, \mathbf{b}^{det}$))
$\mathbf{f}_{fused}$	Fused representation	$\mathbb{R}^{B \times D}$	Residual Contribution Fusion output
λ_{GRL}	GRL intensity	scalar	Gradient reversal coefficient
Output	Class prediction logits	$\mathbb{R}^{B \times C_{cls}}$	Multi-head ensemble output
ε	Label smoothing factor	$[0, 1]$	Smoothing parameter for LSCE
$\mathcal{L}_{LSCE}$	LSCE loss	scalar	Label-Smoothing Cross-Entropy (LSCE)
$\mathcal{L}_{ortho}$	Orthogonal loss	scalar	Cosine + decorrelation across paths
$\mathcal{L}_{WTAL}$	WTAL Loss	scalar	5-component temporal localization loss

discusses conclusions and future research directions.

II. RELATED WORK

Video Transformers and Spatio-Temporal Modeling.

ViT [1] models long-range dependencies via self-attention on patch embeddings. TimeSformer [2] and ViViT [3] decouple spatial and temporal attention, while Video Swin [4] uses hierarchical local windows. MViT [5] and UniFormerV2 [6] improve multi-scale temporal modeling, and VideoMamba [8] applies state space models to video. 3D CNNs such as C3D [31], I3D [32], R(2+1)D [10], and SlowFast [11] jointly process spatial and temporal dimensions. Attention mechanisms including SE-Net [16], CBAM [17], DANet [18], and STCA-Net [14] enhance feature selectivity.

Motion Gating, Fusion, and Person Attention. TSM [12], Non-local [13], GSNet [24], and TEA [25] model temporal dynamics via feature shifting, self-attention, and motion gating. The proposed MGU extends Sobel-based motion encoding with multi-scale bidirectional differencing and boundary detection. Multi-stream approaches [26], [27], [29], [34] and gradient reversal [30] provide complementary baselines for feature diversity. Person localization via Faster R-CNN [19] and ACRN [20] shows the importance of person-centric attention; the proposed module uses YOLO-based detection with Gaussian heatmap weighting.

Weakly-Supervised Temporal Action Localization. WTAL methods [35]–[42] localize frame-level action boundaries using only video-level labels, employing MIL supervision, pseudo-label refinement, and background suppression. The proposed WTAL head recycles MGU features with a 5-component loss, enabling frame-level scoring without additional backbone computation.

III. STCA-ViT NETWORK

A. Overall Architecture

STCA-ViT is a video action recognition framework consisting of six main components: (1) VideoMAE ViT-Base backbone [43], (2) Bidirectional spatio-temporal Cross-Attention (STCA) Module, (3) Sobel-based Motion Gate Unit (MGU), (4) YOLO-Based Person Attention Module [44], (5) Gradient reversal-based Residual Contribution Fusion [30], (6) Multi-Head Ensemble Classifier. An optional WTAL head reuses MGU temporal features to support frame-level action scoring.

The Overall Architecture consists of four parallel pathways. A VideoMAE ViT-Base backbone pretrained on Kinetics-400 extracts D -dimensional spatio-temporal features $\mathbf{b}_{back} \in \mathbb{R}^{B \times D}$ from the input video. Three auxiliary modules independently process the raw video in parallel, each outputting $\mathbf{f}_{stca}, \mathbf{f}_{mgu}, \mathbf{f}_{person} \in \mathbb{R}^{B \times D_m}$. These four features pass through Residual Contribution Fusion and 5-token self-attention refinement before being fed into the Multi-Head Ensemble Classifier. The input video is represented as a 5D tensor:

$$\mathbf{x} \in \mathbb{R}^{B \times C \times T \times H \times W} \quad (1)$$

where B, C, T, H, W denote batch size, number of channels, number of frames, spatial height, and spatial width, respectively. The module embedding dimension D_m defaults to D and can optionally be reduced for memory efficiency.

Fig. 1 illustrates the Overall Architecture. The VideoMAE backbone internally resizes inputs to 224×224 , while auxiliary modules operate at 112×112 resolution to reduce computation. Since D_m can be set independently of D , the complexity of auxiliary modules can be controlled while preserving backbone capacity. The backbone is partially fine-tuned by freezing the early layers and adapting only the later layers

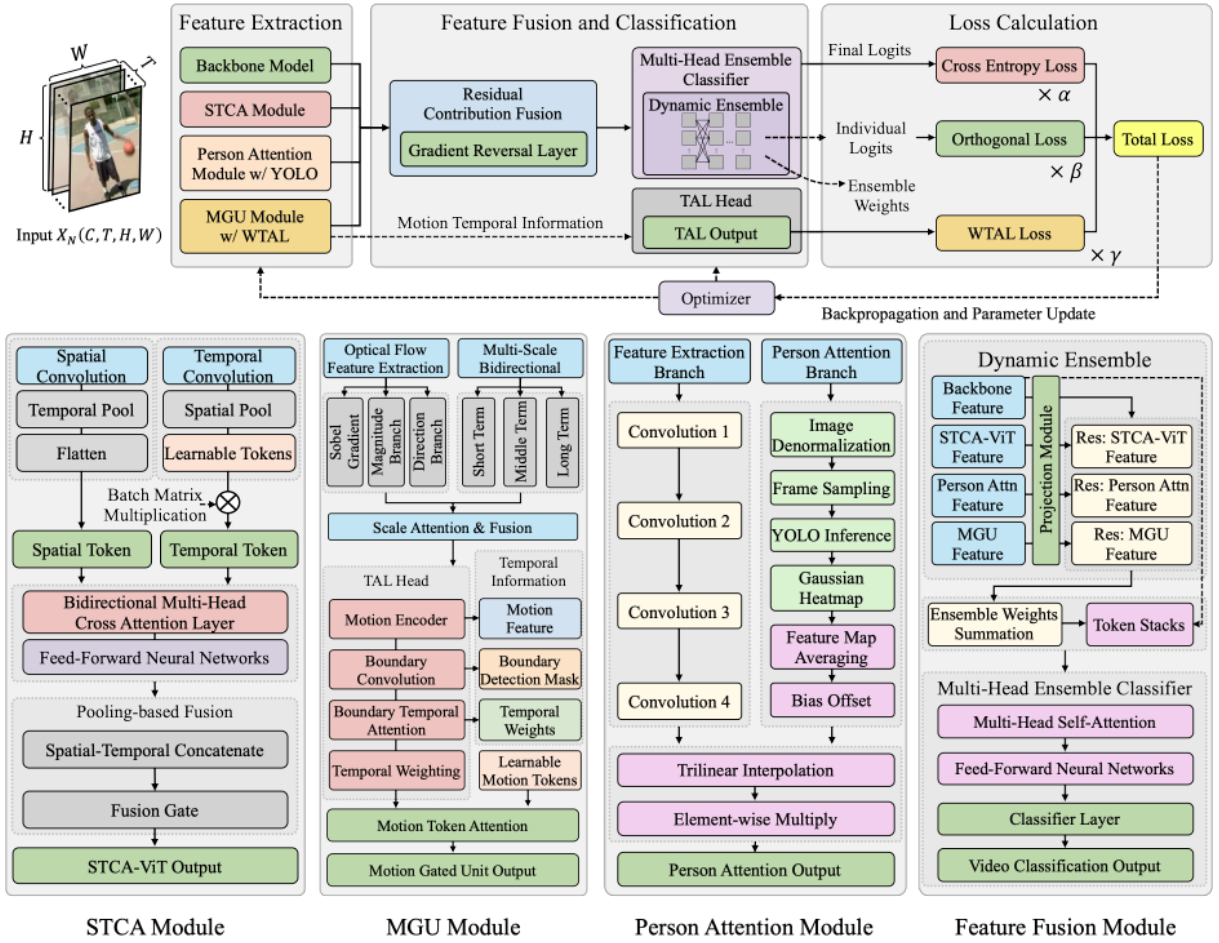


Fig. 1: Overall architecture of the proposed STCA-ViT framework. The input video is processed through four parallel pathways (Backbone, STCA, Person Attention, MGU+WTAL), and the final prediction is generated via Residual Contribution Fusion and the Multi-Head Ensemble Classifier. The internal flow of each module (STCA cross-attention, MGU motion gating, Person Attention, Multi-Head Ensemble Classifier) is included. In the loss computation stage, Cross Entropy Loss($\times\alpha$), Orthogonal Loss($\times\beta$), and WTAL Loss($\times\gamma$) are summed into the Total Loss.

to the target dataset. The additional discriminative power of the auxiliary modules is ensured through orthogonal loss regularization and gradient reversal to guarantee feature diversity.

B. Spatio-Temporal Cross-Attention (STCA) Module

The STCA Module independently extracts spatial tokens and temporal tokens from the raw input $\mathbf{x}$ through hierarchical Conv3d pathways, then aligns the two streams via bidirectional cross-attention. The explicit cross-modal attention combining 3-stage convolutional token extraction with learnable temporal tokens provides richer feature representations than simple patch embedding approaches.

The module consists of four stages as illustrated in Fig. 1 and Fig. 2: (1) Spatial Token Extraction via a 3-stage Conv3d pathway, (2) Temporal Token Extraction via a separate 3-stage Conv3d pathway augmented with learnable tokens, (3) Bidirectional Cross-Attention between the spatial and temporal token sets, and (4) gated pooling-based fusion.

Spatial Token Extraction: A 3-stage Conv3d pathway with a fixed temporal kernel size $k_T=1$ progressively compresses the intra-frame spatial structure, gradually

expanding channels to D_m while reducing spatial resolution. Each stage applies batch normalization and GELU activation, and spatial tokens are produced after temporal averaging (Flat=Flatten, AP=AvgPool, $\text{Conv}^{sp}/\text{Conv}^{tm}$ =3-stage Conv3d spatial/temporal pathway):

$$\mathbf{S}_{tok} = \text{Flat}(\text{AP}_T(\text{Conv}^{sp}(\mathbf{x}))) \in \mathbb{R}^{B \times N_s \times D_m} \quad (2)$$

where $N_s = H' \times W'$ is the number of spatial tokens after downsampling.

Temporal Token Extraction: A separate 3-stage Conv3d pathway with temporal kernels $k=(3, \cdot, \cdot)$ captures inter-frame dynamic characteristics. After spatial averaging, the convolutional temporal features are augmented with N_L learnable tokens $\mathbf{L}_T \in \mathbb{R}^{1 \times N_L \times D_m}$ via scaled dot-product attention:

$$\mathbf{T}_{conv} = \text{Flat}(\text{AP}_{HW}(\text{Conv}^{tm}(\mathbf{x}))) \in \mathbb{R}^{B \times T' \times D_m} \quad (3)$$

$$\mathbf{L}'_T = \mathbf{L}_T + \text{Attn}(\mathbf{L}_T, \text{Proj}(\mathbf{T}_{conv}), \text{Proj}(\mathbf{T}_{conv})) \quad (4)$$

$$\mathbf{T}_{tok} = [\mathbf{T}_{conv}; \mathbf{L}'_T] \in \mathbb{R}^{B \times (T' + N_L) \times D_m} \quad (5)$$

where $[\cdot; \cdot]$ denotes concatenation along the token dimension. The $N_L=32$ learnable tokens function as free temporal

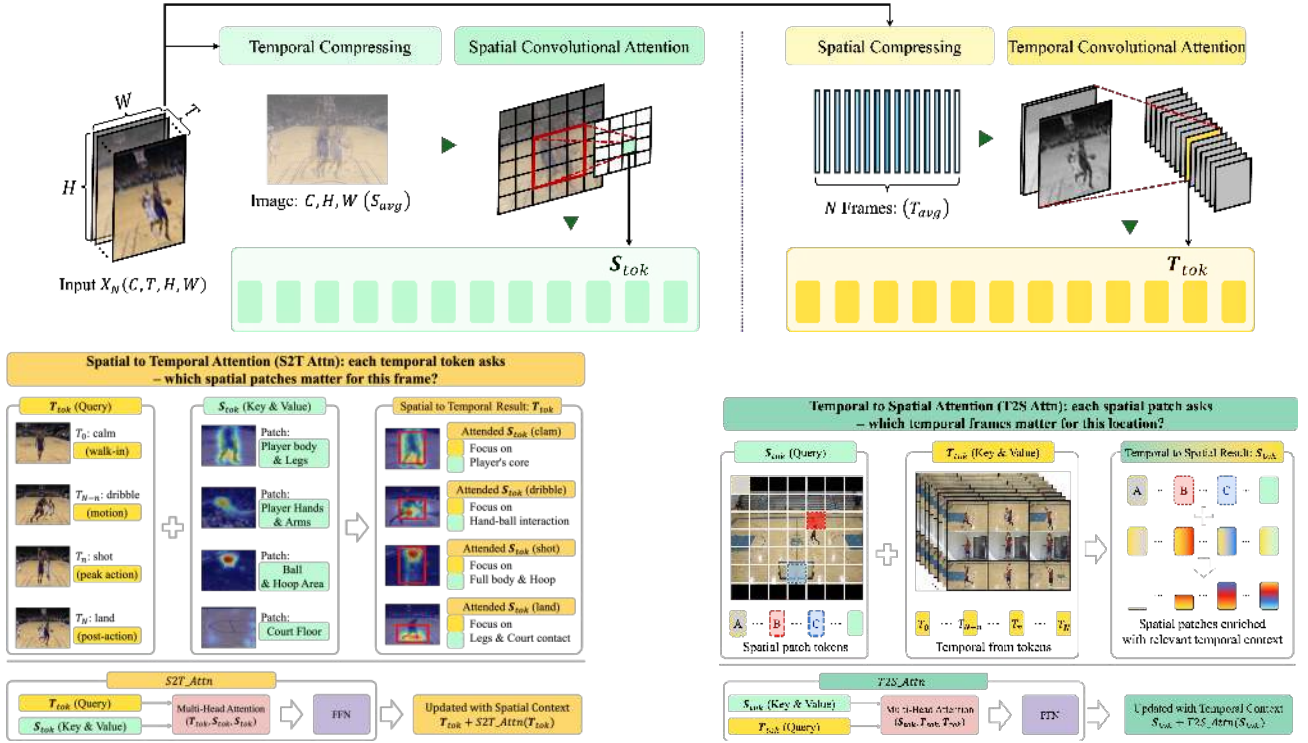


Fig. 2: Token extraction and bidirectional cross-attention in the STCA Module. (Top) Extraction process for spatial tokens (S_{tok}) and temporal tokens (T_{tok}): (Left) Temporal Compressing followed by Spatial Convolutional Attention to produce S_{tok} ; (Right) Spatial Compressing followed by Temporal Convolutional Attention to produce T_{tok} . (Bottom-left) Spatial-to-Temporal Attention (S2T Attn): each temporal token (T_{tok} , Query) selects the spatially important patches for the corresponding frame from the spatial tokens (S_{tok} , Key/Value), enriching temporal context with spatial information. (Bottom-right) Temporal-to-Spatial Attention (T2S Attn): each spatial patch (S_{tok} , Query) selects the temporally important frames for the corresponding location from the temporal frame sequence (T_{tok} , Key/Value), enriching spatial representations with temporal context.

prototypes that capture global and abstract temporal patterns beyond the limitations of the convolutional receptive field, enhancing query diversity in cross-attention.

Bidirectional Cross-Attention: The spatial and temporal tokens interact through L bidirectional cross-attention layers. Each layer applies pre-normalization cross-attention with residual connections ($\tilde{T} \equiv \text{LN}(T_{tok}^{(l-1)})$, $\tilde{S} \equiv \text{LN}(S_{tok}^{(l-1)})$):

$$T_{tok}^{(l)} \leftarrow \text{LN}\left(T_{tok}^{(l-1)} + \text{DP}\left(\text{MHA}(\tilde{T}, \tilde{S}, \tilde{S})\right)\right) \quad (6)$$

$$S_{tok}^{(l)} \leftarrow \text{LN}\left(S_{tok}^{(l-1)} + \text{DP}\left(\text{MHA}(\tilde{S}, \tilde{T}, \tilde{T})\right)\right) \quad (7)$$

where $\text{DP}(\cdot)$ denotes DropPath regularization. Each stream is then refined by a feed-forward network (FFN) with pre-normalization and residual connections:

$$\begin{aligned} S_{tok}^{(l)} &\leftarrow S_{tok}^{(l)} + \text{DP}\left(\text{FFN}(\text{LN}(S_{tok}^{(l)}))\right) \\ T_{tok}^{(l)} &\leftarrow T_{tok}^{(l)} + \text{DP}\left(\text{FFN}(\text{LN}(T_{tok}^{(l)}))\right) \end{aligned} \quad (8)$$

The FFN is a 2-layer structure (hidden dimension $4D_m$) with GELU activation and dropout. The bidirectional design allows temporal tokens to reference spatial structure while spatial tokens reflect temporal dynamics, capturing cross-modal dependencies that are difficult to express through element-wise fusion.

Gated Fusion: Average pooling is applied to the final tokens of each stream to obtain $\bar{s} = \text{AvgPool}(S_{tok}^{(L)})$, $\bar{t} =$

$$\text{AvgPool}(T_{tok}^{(L)}):$$

$$\mathbf{w} = \text{Softmax}(\text{FC}([\bar{s}; \bar{t}])) \in \mathbb{R}^{B \times 2} \quad (9)$$

$$\mathbf{f}_{stca} = \text{Proj}(w_1 \cdot \bar{s} + w_2 \cdot \bar{t}) \in \mathbb{R}^{B \times D_m} \quad (10)$$

The gating weights dynamically adjust the spatial and temporal contributions for each sample, emphasizing the more informative stream.

C. Motion Gate Unit (MGU)

The MGU extracts explicit motion features directly from the raw input through Sobel-based gradient computation, Multi-Scale Bidirectional temporal differencing, and Motion Boundary Detection. The overall structure is illustrated in Fig. 3.

Sobel-Based Gradient Extraction: Spatial gradients are computed using depthwise Sobel filters initialized with horizontal and vertical kernels:

$$\mathbf{G}_x, \mathbf{G}_y = \text{SobelConv3d}(\mathbf{x}) \in \mathbb{R}^{B \times 2C \times T \times H \times W} \quad (11)$$

The temporal difference $\Delta|\nabla|_t = |\nabla|_t - |\nabla|_{t-1}$ captures spatial changes in moving object outlines, detecting structural motion unlike simple frame differencing which only observes pixel brightness. Computation is significantly lower than optical flow estimation. Motion magnitude and directional

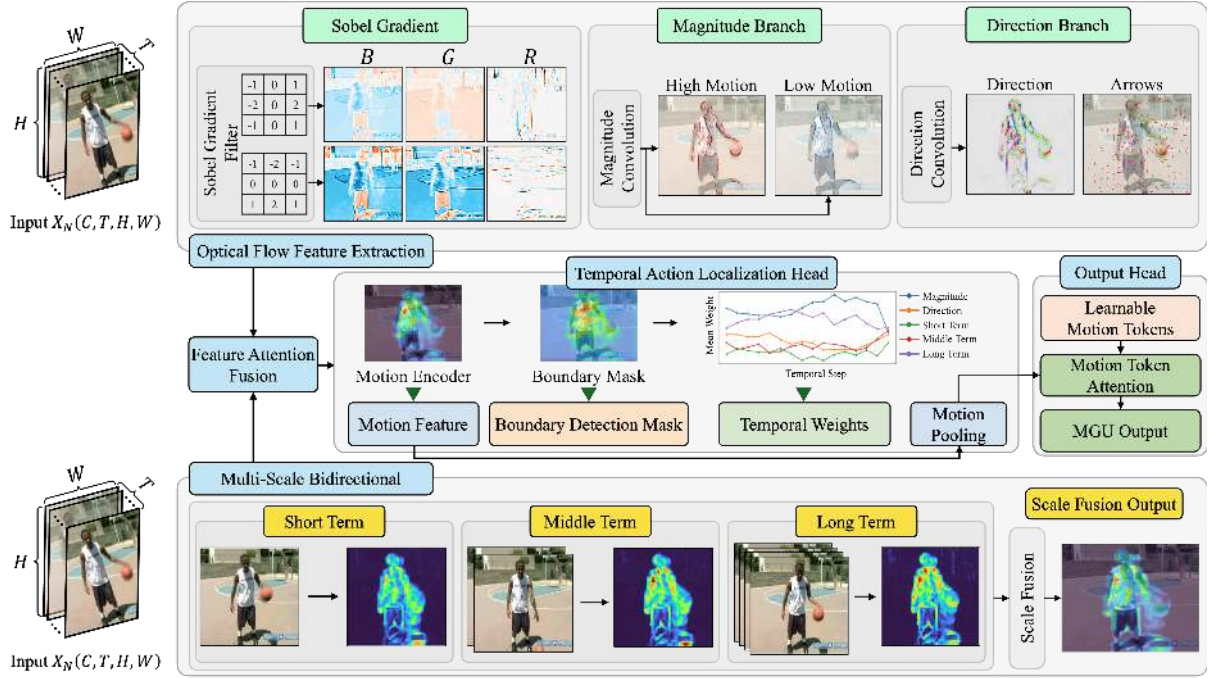


Fig. 3: Detailed architecture of the MGU module. (Top) Optical flow feature extraction via Sobel Gradient filters and Magnitude/Direction Branch processing; (Middle) Motion Encoder, Boundary Mask, and Temporal Weights computation and TAL Head structure; (Bottom) Multi-Scale Bidirectional processing (Short/Middle/Long Term) and Scale Fusion output.

features are obtained from the temporal difference of these gradients:

$$\begin{aligned} \mathbf{F}_{mag} &= \text{Conv3d}(|\nabla|_t - |\nabla|_{t-1}) \in \mathbb{R}^{B \times C_m \times T \times H \times W} \\ \mathbf{F}_{dir} &= \text{Conv3d}([\Delta \mathbf{G}_x; \Delta \mathbf{G}_y]) \in \mathbb{R}^{B \times C_m \times T \times H \times W} \end{aligned} \quad (12)$$

where $|\nabla| = \sqrt{\mathbf{G}_x^2 + \mathbf{G}_y^2}$ and Δ denotes temporal difference.

Multi-Scale Bidirectional Motion: Forward and backward temporal differences at scales $s \in \{1, 2, 4\}$ capture motion at different temporal granularities. $s=1$ handles short-term movement, $s=2$ handles medium-term changes, and $s=4$ handles long-term patterns, while bidirectional differences handle asymmetric motion symmetrically:

$$\mathbf{F}_s = \text{Conv3d}([\mathbf{x}_{t+s} - \mathbf{x}_t; \mathbf{x}_{t-s} - \mathbf{x}_t]) \in \mathbb{R}^{B \times C_m \times T \times H \times W} \quad (13)$$

Replication padding is applied at temporal boundaries.

Scale Attention and Fusion: All motion features (magnitude, direction, multi-scale) are concatenated and weighted by learned scale attention:

$$\mathbf{F}_{mot} = \text{Conv3d}\left(\sum_i \sigma_i \cdot \mathbf{F}_i\right) \in \mathbb{R}^{B \times C_f \times T \times H' \times W'} \quad (14)$$

where $\sigma_i = \text{Softmax}(\text{Conv1d}([\mathbf{F}_{mag}; \mathbf{F}_{dir}; \mathbf{F}_{s1}; \dots; \mathbf{F}_{s3}]))_i$ computes per-scale importance.

Motion Encoder: A 3-stage Conv3d encoder maps the fused motion features to the embedding dimension:

$$\mathbf{M}_{enc} = \text{Conv3d}_{3\text{-stage}}(\mathbf{F}_{mot}) \in \mathbb{R}^{B \times D_m \times T' \times H'' \times W''} \quad (15)$$

Progressive channel expansion ($C_f \rightarrow \dots \rightarrow D_m$) and spatio-temporal downsampling are applied.

Motion Boundary Detection: A 2-layer boundary detection branch generates a sigmoid mask highlighting temporal motion boundaries:

$$\mathbf{H} = \text{GELU}\left(\text{BN}\left(\text{Conv3d}_{1 \times 3 \times 3}^{D_m \rightarrow D_m/4}(\mathbf{M}_{enc})\right)\right)$$

$$\mathbf{B}_{mask} = \sigma(\text{Conv3d}_{1 \times 1}(\mathbf{H})) \in \mathbb{R}^{B \times 1 \times T' \times H'' \times W''} \quad (16)$$

The encoded features are enhanced using the mask: $\mathbf{M}_{enc} \leftarrow \mathbf{M}_{enc} \odot (1 + \mathbf{B}_{mask})$. Since $\mathbf{B}_{mask} \in [0, 1]$, the multiplier $(1 + \mathbf{B}_{mask}) \in [1, 2]$ preserves the original intensity in regions without boundaries, while locations with strong motion boundaries are emphasized up to twice, making action start/end transition points prominent in the feature map.

Temporal Attention: After spatial global average pooling, temporal attention weights are computed via a 2-layer projection with GELU:

$$\begin{aligned} \mathbf{p} &= \text{GELU}(\text{FC}_{D_m \rightarrow D_m/4}(\text{AP}_{HW}(\mathbf{M}_{enc}))) \\ \mathbf{w}_{temp} &= \text{Softmax}(\text{FC}_{D_m/4 \rightarrow 1}(\mathbf{p})) \in \mathbb{R}^{B \times 1 \times T' \times 1 \times 1} \end{aligned} \quad (17)$$

Learnable Motion Token: A learnable motion token $\mathbf{m}_{tok} \in \mathbb{R}^{1 \times D_m}$ is used as a query in multi-head attention over temporally-weighted motion features to produce a compact global motion representation:

$$\mathbf{M}' = \text{MHA}(\text{Flat}(\mathbf{M}_{enc} \odot \mathbf{w}_{temp}), \mathbf{m}_{tok}, \mathbf{m}_{tok}) \quad (18)$$

$$\mathbf{f}_{mgu} = \text{Proj}(\text{AvgPool}(\mathbf{M}')) \in \mathbb{R}^{B \times D_m} \quad (19)$$

In WTAL mode, the MGU additionally returns the motion features $\mathbf{M}_{enc}$, boundary mask $\mathbf{B}_{mask}$, and temporal weights $\mathbf{w}_{temp}$ to be passed to the WTAL head (Section III-G).

D. Person Attention Module

The Person Attention Module combines YOLOv8 [44]-based person detection with Gaussian heatmap weighting to emphasize person-relevant regions. A 4-stage Conv3d pathway extracts hierarchical spatial features from the input, expanding channels to D_m :

$$\mathbf{F}_{person} = \text{Conv3d}_{4\text{-stage}}(\mathbf{x}) \in \mathbb{R}^{B \times D_m \times T \times H' \times W'} \quad (20)$$

For each batch element, k uniformly sampled frames are passed to a lazily initialized YOLOv8-nano detector. A Gaussian attention map is generated at each detected person center (c_x, c_y) , where σ is proportional to the bounding box size:

$$G(x, y) = \exp\left(-\frac{(x - c_x)^2 + (y - c_y)^2}{2\sigma^2}\right) \quad (21)$$

Per-frame heatmaps are merged and averaged to form $\mathbf{A}_{person} \in \mathbb{R}^{B \times 1 \times T \times H \times W}$, with a minimum offset ensuring non-zero attention when no person is detected. The attention tensor is resized via trilinear interpolation and applied element-wise before pooling:

$$\mathbf{f}_{person} = \text{Proj}(\text{AvgPool}(\mathbf{F}_{person} \odot \text{Interp}(\mathbf{A}_{person}))) \in \mathbb{R}^{B \times D_m} \quad (22)$$

During training, detector inputs are detached to block gradient flow through the detection branch.

E. Residual Contribution Fusion and Multi-Head Ensemble Classifier

Residual Contribution Fusion. The four features $\mathbf{b}_{back}$, $\mathbf{f}_{stca}$, $\mathbf{f}_{mgu}$, $\mathbf{f}_{person}$ are integrated via Residual Contribution Fusion. Each auxiliary feature first passes through GRL [30], which acts as identity during the forward pass but negates gradients during backpropagation ($\nabla \rightarrow -\lambda_{GRL} \nabla$), encouraging each module to explore complementary representation spaces:

$$\text{GRL}_\lambda(\mathbf{x}) = \mathbf{x} \text{ (forward), } \nabla \rightarrow -\lambda \nabla \text{ (backward)} \quad (23)$$

Per-module residuals are computed by concatenating each GRL-processed feature with a stop-gradient backbone copy and passing through a 2-layer FC block (GELU + LN):

$$\mathbf{r}_i = \text{FC}([\mathbf{f}_i; \mathbf{b}^{\text{det}}]) \in \mathbb{R}^{B \times D}, \quad i \in \{\text{stca}, \text{mgu}, \text{person}\} \quad (24)$$

The fused representation adds sigmoid-gated residuals to the backbone projection:

$$\mathbf{f}_{fused} = \mathbf{b} + \sigma(\alpha)\mathbf{r}_{stca} + \sigma(\beta)\mathbf{r}_{mgu} + \sigma(\gamma)\mathbf{r}_{person} \quad (25)$$

where α, β, γ are learnable scalars. The five tokens $\{\mathbf{b}, \mathbf{f}_{fused}, \mathbf{r}_{stca}, \mathbf{r}_{mgu}, \mathbf{r}_{person}\}$ are then refined via a single-layer multi-head self-attention block (FFN hidden dim $4D$), and the fused token $\mathbf{f}_{out} = \text{Proj}(\hat{\mathbf{Z}}_{[1]})$ is used for classification.

Multi-Head Ensemble Classifier. Five parallel classification heads produce per-module logits $\mathbf{y}_k$, each following $\text{FC} \rightarrow \text{LN} \rightarrow \text{GELU} \rightarrow \text{Dropout} \rightarrow \text{FC}$.

Input-dependent dynamic ensemble weights are computed from the concatenation of all module features:

$$\mathbf{w} = \text{Softmax}(\text{FC}([\mathbf{b}_{back}; \mathbf{f}_{stca}; \mathbf{f}_{mgu}; \mathbf{f}_{person}; \mathbf{f}_{fused}])) \in \mathbb{R}^{B \times 5} \quad (26)$$

$$\text{Output} = \sum_{k=1}^5 w_k \cdot \mathbf{y}_k \in \mathbb{R}^{B \times C_{cls}} \quad (27)$$

Static initialization weights $[0.2, 0.2, 0.2, 0.1, 0.3]$ stabilize early training, after which dynamic weighting adapts per sample. The entire model is optimized end-to-end.

F. Loss Function

Label-Smoothing Cross-Entropy (LSCE). LSCE relaxes the one-hot targets with a smoothing parameter $\varepsilon \in [0, 1]$ to suppress overconfidence. The per-class target probability for the ground-truth label y is:

$$t_c = \begin{cases} 1 - \varepsilon & \text{if } c = y \\ \frac{\varepsilon}{C_{cls} - 1} & \text{otherwise} \end{cases} \quad (28)$$

The loss is computed as the cross-entropy between the smoothed distribution and the predicted distribution:

$$\mathcal{L}_{LSCE} = -(1 - \varepsilon) \log(\hat{p}_y) - \frac{\varepsilon}{C_{cls} - 1} \sum_{c \neq y} \log(\hat{p}_c) \quad (29)$$

where $\hat{p}_c = \text{softmax}(\mathbf{p})_c$. The first term encourages focus on the correct class, while the second term penalizes overconfidence on incorrect classes. Smoothing suppresses the softmax output from converging to 1.0, reducing the Expected Calibration Error (ECE) and improving generalization.

Orthogonal Loss and Gradient Reversal. Two regularization techniques are applied to encourage auxiliary modules to learn complementary features. The GRL (Eq. (23)) linearly anneals λ_{GRL} from 0 to the target value to progressively increase diversity pressure. The orthogonal loss combines cosine similarity minimization with cross-correlation decorrelation for all pairs of the projected pathway features $\{\mathbf{b}, \mathbf{c}, \mathbf{m}, \mathbf{p}\}$:

$$\mathcal{L}_{ortho} = \sum_{(i,j) \in \mathcal{P}} \left[\left(\frac{\mathbf{f}_i^\top \mathbf{f}_j}{\|\mathbf{f}_i\| \|\mathbf{f}_j\|} \right)^2 + \lambda_d \|\mathbf{C}_{ij}\|_F^2 \right] \quad (30)$$

where $\mathbf{C}_{ij} = \frac{1}{B} \tilde{\mathbf{f}}_i^\top \tilde{\mathbf{f}}_j$ is the cross-correlation matrix of standardized features, and $\lambda_d = 0.005$. The cosine term suppresses directional similarity of feature vectors (macro-level redundancy), while the cross-correlation term suppresses dimension-wise collinearity (micro-level correlation).

G. Weakly-Supervised Temporal Action Localization Head

The optional WTAL head reuses the temporal features from the MGU to generate frame-level action scores without temporal annotations. Since $\mathbf{M}_{enc}$, $\mathbf{B}_{mask}$, and $\mathbf{w}_{temp}$ already computed by the MGU can be directly used to distinguish action frames from background frames, they are well-suited as inputs to the WTAL head without a separate localization network. The head receives three inputs from the MGU when

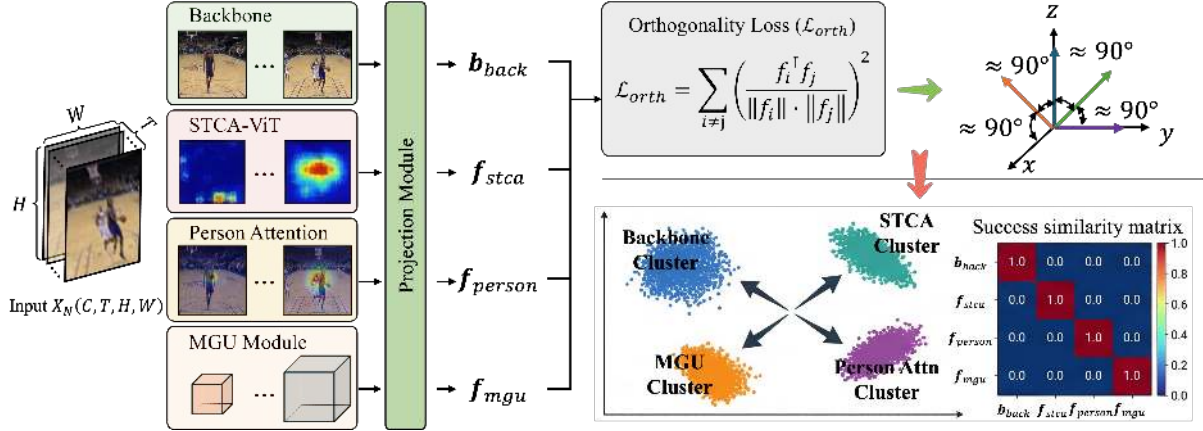


Fig. 4: Structure of the orthogonal loss ($\mathcal{L}_{orth}$). Each module feature (Backbone, STCA-ViT, Person Attention, MGU) is projected via a Projection Module, and cosine similarity minimization is applied to all pairs. The similarity matrix on the right shows the ideal state after training, where features across modules are mutually orthogonal (decorrelated).

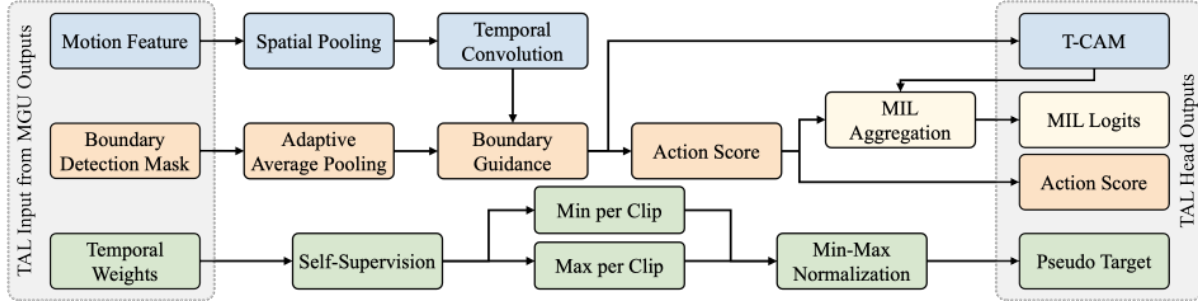


Fig. 5: Internal structure of the TAL Head. Taking as input the MGU outputs—Motion Feature, Boundary Detection Mask, and Temporal Weights—it passes through Spatial Pooling, Temporal Convolution, and Boundary Guidance to output T-CAM, MIL Logits, and Action Score. Temporal Weights are converted to Pseudo Targets via Min-Max Normalization through the Self-Supervision pathway, providing a weakly-supervised learning signal.

return_temporal=True: $\mathbf{M}_{enc} \in \mathbb{R}^{B \times D_m \times T' \times H'' \times W''}$, $\mathbf{B}_{mask}$, and $\mathbf{w}_{temp}$.

Architecture: The motion features are spatially pooled to obtain a temporal feature sequence:

$$\mathbf{h} = \text{AvgPool}_{HW}(\mathbf{M}_{enc}) \in \mathbb{R}^{B \times D_m \times T'} \quad (31)$$

Depthwise separable temporal convolution with a residual connection refines the temporal features:

$$\mathbf{h} \leftarrow \text{LN}(\mathbf{h} + \text{DWSepConv1d}(\mathbf{h})) \quad (32)$$

DWSepConv1d consists of a depthwise Conv1d (kernel 3, groups= D_m) followed by a pointwise Conv1d, each with batch normalization and GELU.

The features are enhanced with the boundary mask:

$$\mathbf{h} \leftarrow \mathbf{h} \odot (1 + \text{AvgPool}_{HW}(\mathbf{B}_{mask})) \quad (33)$$

The head produces two outputs: a T-CAM through a 1×1 convolution with dropout, and per-frame action scores through a 2-layer branch:

$$\mathbf{T}_{cam} = \text{Conv1d}_{1 \times 1}(\text{Dropout}(\mathbf{h})) \in \mathbb{R}^{B \times C_{cls} \times T'} \quad (34)$$

$$\mathbf{a} = \sigma\left(\text{Conv1d}_{1 \times 1}\left(\text{GELU}\left(\text{Conv1d}_{1 \times 1}^{D_m \rightarrow D_m/4}(\mathbf{h})\right)\right)\right) \in \mathbb{R}^{B \times T'} \quad (35)$$

MIL Aggregation: Clip-level logits are obtained via MIL pooling [36]. The T-CAM is weighted by normalized action scores:

$$\hat{\mathbf{a}} = \frac{\mathbf{a}}{\sum_t a_t + \epsilon}, \quad \mathbf{y}_{mil} = \mathbf{T}_{cam} \cdot \hat{\mathbf{a}}^\top \in \mathbb{R}^{B \times C_{cls}} \quad (36)$$

Pseudo Target Generation: $\mathbf{w}_{temp}$ is min-max normalized to $[0, 1]$ per sample and used as pseudo supervision targets for the action scores. This provides a self-supervised learning signal without temporal annotations.

The WTAL head introduces negligible additional parameters and computational overhead compared to the full model, and reuses features already computed by the MGU.

H. WTAL Loss

The WTAL Loss consists of 5 terms that train the temporal localization head using only clip-level labels:

$$\mathcal{L}_{WTAL} = \lambda_1 \mathcal{L}_{mil} + \lambda_2 \mathcal{L}_{pseudo} + \lambda_3 \mathcal{L}_{sparse} + \lambda_4 \mathcal{L}_{smooth} + \lambda_5 \mathcal{L}_{bg} \quad (37)$$

(1) **MIL Classification Loss** (λ_1): Cross-entropy between the MIL-aggregated logits and the clip-level labels, serving as the primary supervision signal for the WTAL head: $\mathcal{L}_{mil} = \text{CE}(\mathbf{y}_{mil}, \mathbf{y})$.

(2) **Pseudo-Label BCE** (λ_2): Binary cross-entropy between the predicted action scores and the MGU pseudo targets:

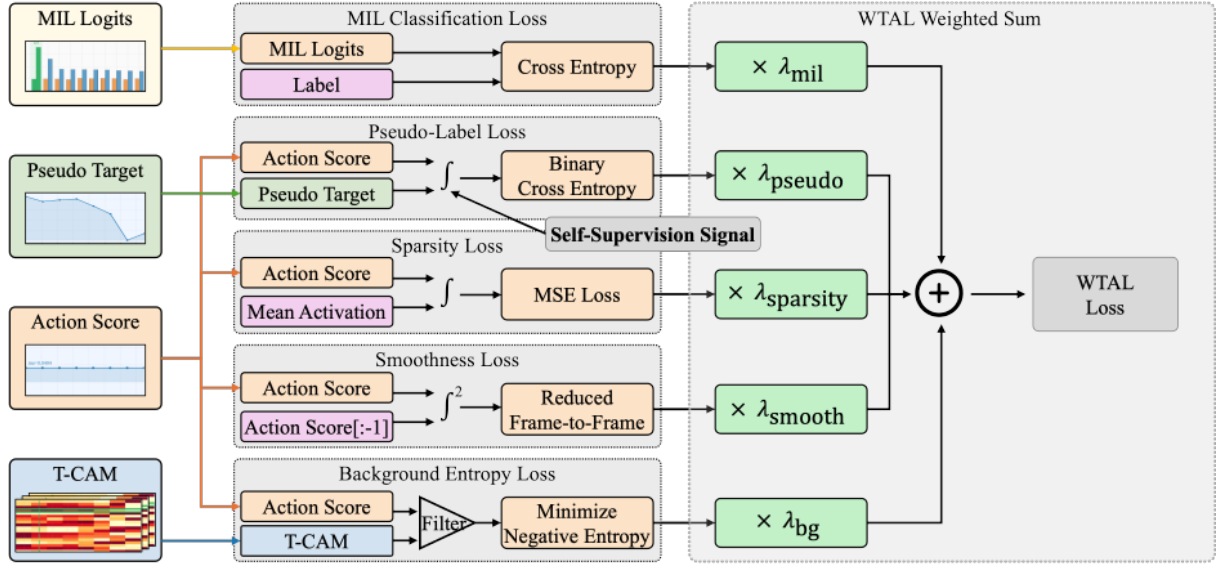


Fig. 6: The 5 components of the WTAL Loss. Taking MIL Logits, Pseudo Target, Action Score, and T-CAM as inputs, it computes (1) MIL Classification Loss, (2) Pseudo-Label Loss, (3) Sparsity Loss, (4) Smoothness Loss, and (5) Background Entropy Loss, and sums them with respective weights (λ_{mil} , λ_{pseudo} , $\lambda_{sparsity}$, λ_{smooth} , λ_{bg}) to produce the final WTAL Loss.

$\mathcal{L}_{pseudo} = \text{BCE}(\mathbf{a}, \hat{\mathbf{w}}_{temp}^{det})$, where $\hat{\mathbf{w}}_{temp}^{det}$ is the detached normalized temporal attention.

(3) **Sparsity Regularization** (λ_3): Drives the mean action activation toward a target value $\tau_s=0.5$ to prevent convergence to all-foreground or all-background solutions: $\mathcal{L}_{sparse} = \text{MSE}(\bar{a}, \tau_s)$.

(4) **Temporal Smoothness** (λ_4): Suppresses abrupt oscillations in frame-level predictions to maintain smooth transitions: $\mathcal{L}_{smooth} = \frac{1}{T'-1} \sum_t (a_t - a_{t-1})^2$.

(5) **Background Entropy** (λ_5): Maximizes the class distribution entropy for background frames where $a_t < \tau_b=0.5$, suppressing confident activation of the T-CAM toward specific classes in background segments: $\mathcal{L}_{bg} = \log(C_{cls}) - \mathcal{H}(\bar{\mathbf{p}}_{bg})$.

The WTAL Loss is activated after a warmup of E_{wtal} epochs, with the loss weight linearly increasing from 0 to λ_{wtal} .

IV. EXPERIMENTS

A. Datasets and Implementation Details

TABLE II: Experimental training environment and configuration.

Item	Specification
Hardware	NVIDIA H100 (80 GB HBM3), Intel Xeon Silver 4514Y, 252 GB RAM
Software	PyTorch 2.1, CUDA/cuDNN, Python 3.10, Weights & Biases
Datasets	HMDB-51, UCF-101, Kinetics-400, SSv2 (official splits)
Input	16-frame clips; 224×224 (backbone), 112×112 (auxiliary)
Augmentation	RandAugment (n=2, m=9), CutMix ($\alpha=1.0$), MixUp ($\alpha=0.4$), Random Erasing (p=0.25)
Optimizer	AdamW (wd = 0.05); backbone lr 1×10^{-6} (bottom 6 layers frozen)
Precision	Mixed precision (fp16, AMP), gradient accumulation, EMA (decay 0.9999)
Regularization	Dropout = 0.3, gradient clipping (max-norm = 1.0)

Experiments are conducted on four standard benchmarks: UCF-101 (101 classes, 13,320 clips, 3-split), HMDB-51 (51 classes, 6,766 clips, 3-split), Kinetics-400 (400 classes, ~300K videos), and SSv2 (174 classes, ~220K videos, official train/val split). Videos are sampled as 16-frame clips with a temporal stride of 4 (UCF-101, Kinetics-400) or 2 (HMDB-51, SSv2). Standard augmentations (RandAugment, CutMix, MixUp, Random Erasing) are applied during training, with Flip TTA at test time (6-view for Kinetics-400).

The model is trained with AdamW (weight decay 0.05), a base lr of 1×10^{-4} (SSv2: 5×10^{-5}), and backbone lr of 1×10^{-6} with the bottom 6 of 12 layers frozen. A linear warmup followed by cosine annealing is used as the scheduler. Loss weights are $\lambda_{ortho}=0.1$, $\lambda_{GRL}=0.1$, and $\lambda_{wtal}=0.3$ (activated after warmup). Training uses mixed-precision (fp16) with EMA (decay 0.9999), dropout (0.3), and gradient clipping (max-norm 1.0). The full setup is summarized in Table II.

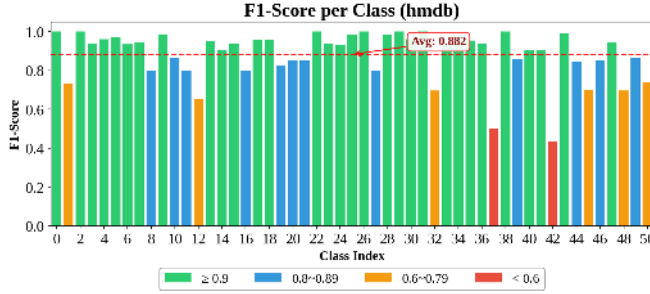
B. Performance Comparison with Baselines

Table III compares Top-1 accuracy across four benchmarks alongside computational efficiency metrics. STCA-ViT uses the VideoMAE ViT-Base backbone and the full architecture (110M parameters), measured with Flip TTA for HMDB-51/UCF-101/SSv2, and 3-clip + Flip TTA for Kinetics-400.

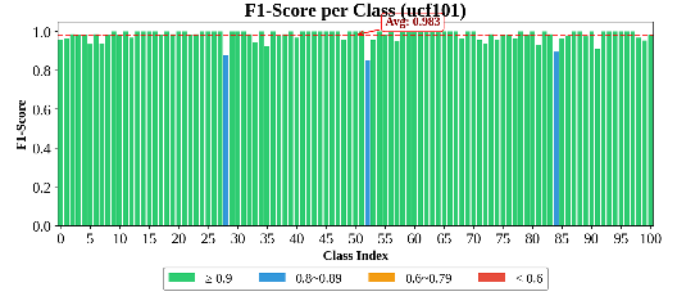
CNN-based models (TSM, SlowFast, TEA) achieve sub-10ms latency but cap at 73–80% on HMDB-51/Kinetics-400. Transformer models (TimeSformer, ViViT, Video Swin) improve accuracy at significantly higher cost. VideoMAE V2 (ViT-g) requires >1B parameters and >500 GFLOPs to achieve 86.08% on HMDB-51. STCA-ViT achieves 90.46% on HMDB-51 (+4.38%p), 98.29% on UCF-101, 79.44% on Kinetics-400, and 65.01% on SSv2 with 149.2M parameters and 355 GFLOPs (28.2 ms/clip, 663 MB GPU memory on H100) — $6.7 \times$ fewer parameters than VideoMAE V2 ViT-g.

TABLE III: Performance and efficiency comparison of action recognition models on major benchmarks. Accuracy is reported as Top-1 (%) for UCF-101 and HMDB-51; K-400 and SSv2 are reported as Top-1/Top-5 (%). GFLOPs are measured on 16-frame video clips (224 resolution), and parameter counts are in millions. Latency is inference latency (ms/clip) measured on an NVIDIA H100 GPU with batch size 1. GPU Mem is peak activation memory during inference (MB). “–” indicates not reported.

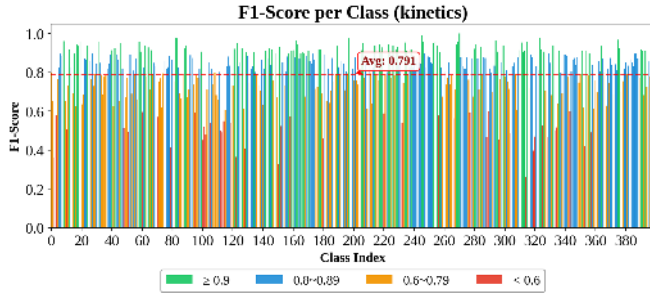
Model	Year	UCF-101	HMDB-51	K-400	SSv2	GFLOPs	Params (M)	Latency (ms)	GPU Mem (MB)
R(2+1)D [10]	2018	96.8	74.5	72.0/–	–	152	63	9.0	580
Non-local NN [13]	2018	88–90	62–66	76.5/93.3	–	359	35	–	–
TSM [12]	2019	95.9	73.5	74.7/91.2	63.4/88.5	33	24	4.4	217
SlowFast [11]	2019	~96	~73	79.8/92.6	61.9/–	213	34	10.2	218
GSNet [24]	2020	96.8	74.5	–	–	33	24	–	–
TEA [25]	2020	96.9	73.3	76.1/92.5	64.6/89.9	34	24	–	–
TimeSformer [2]	2021	–	–	80.7/93.7	62.5/–	196	121	36.0	710
ViViT [3]	2021	–	–	81.3/93.8	65.9/89.9	398	310	43.4	630
MViT [5]	2021	–	–	78.4/93.5	64.7/89.2	70–170	36–53	25.2	434
Video Swin [4]	2022	–	–	84.9/94.6	69.6/92.7	88–200	88	–	–
VideoMAE V2 [45]	2023	91.3	62.6	80.0/–	70.6/–	~200	~200	17.6	434
VideoMAE V2 (ViT-g) [45]	2023	99.05	86.08	90.0/98.1	77.0/95.9	>500	>1000	–	–
VideoMamba [8]	2024	–	68.6	77.7/95.4	64.2/91.4	Low	74	–	–
STCA-ViT (Ours)	2026	98.29	90.46	79.44/94.25	65.01/91.22	355	149.2	28.2	663



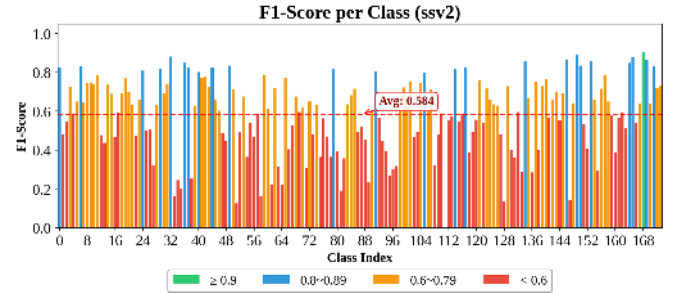
(a) HMDB-51 (Macro F1 = 0.8874)



(b) UCF-101 (Macro F1 = 0.9827)



(c) Kinetics-400 (Macro F1 = 0.7913)



(d) SSv2 (Macro F1 = 0.5845)

Fig. 7: Per-class F1 score distributions on four benchmarks. (a) HMDB-51: most classes achieve high F1, with lower scores for visually similar actions (e.g., *stand*, *sit*). (b) UCF-101: most classes exceed 0.95, with many achieving F1=1.0. (c) Kinetics-400: wide performance spread due to 400-class diversity. (d) SSv2: lower overall F1 reflecting the fine-grained temporal reasoning challenge.

C. Evaluation on Four Benchmarks

STCA-ViT is evaluated on HMDB-51, UCF-101, Kinetics-400, and SSv2, which differ in scale, action diversity, visual ambiguity, and temporal reasoning complexity. We report average accuracy, macro F1 score, and per-class F1 distribution.

1) *HMDB-51*: HMDB-51 features 51 person-centric action classes where all three auxiliary modules operate as intended: YOLO reliably detects subjects, MGU captures limb motion via Sobel gradients, and STCA models spatio-temporal evolution via bidirectional cross-attention. STCA-ViT achieves Top-1 90.46% and macro F1 0.8874, outperforming VideoMAE V2 ViT-g (86.08%) by +4.38%p.

Classes with weak motion cues (*stand*, *sit*) and high visual similarity (*fall_floor*, 67%) are the primary sources of error (Fig. 7(a)).

2) *UCF-101*: UCF-101 comprises 101 sport and daily-activity classes with clear person movements. STCA-ViT achieves 98.29% Top-1 and macro F1 0.9827; over 60 classes record 100% accuracy. Since the backbone-only baseline (A0) already achieves 86.98%, the final gain reflects complementary contributions from the auxiliary modules rather than the backbone alone. Crowd scenes (*MilitaryParade*, 74.07%) reduce person attention focus; ECE 0.0888 confirms calibration quality.

TABLE IV: Performance of the proposed STCA-ViT model across four benchmarks.

Metric	HMDB-51	UCF-101	K-400	SSv2
Top-1 Accuracy (%)	90.46	98.29	79.44	65.01
Top-5 Accuracy (%)	98.52	99.97	94.25	91.22
Macro Precision	0.8809	0.9846	0.8017	0.6038
Macro Recall	0.8874	0.9821	0.7937	0.5881
Macro F1	0.8820	0.9827	0.7913	0.5845
Weighted Precision	0.9020	0.9838	0.8023	0.6527
Weighted Recall	0.8988	0.9829	0.7944	0.6501
Weighted F1	0.8984	0.9827	0.7920	0.6440
ECE	0.0581	0.0888	0.1064	0.0198
Test Samples	1,551	3,160	38,676	24,777

3) *Kinetics-400*: Kinetics-400 covers 400 diverse action classes including non-person-centric scenes. STCA-ViT achieves 79.44% Top-1 and macro F1 0.7913, approximately 0.56%p below VideoMAE V2. Person-centric classes benefit from YOLO attention, while static or non-person scenes (*answering questions*, 31.25%) yield weak module signals (Fig. 7(c)).

4) *Something-Something-v2*: SSv2 focuses on fine-grained hand-object interactions, where YOLO attention and Sobel-based MGU are both limited: YOLO fails on hand-only clips and MGU lacks directional encoding. STCA-ViT achieves 65.01% Top-1 and macro F1 0.5845, a gap of -5.59% vs. VideoMAE V2 (70.6%). Direction-aware encoding and hand-specialized attention remain as future work; ECE 0.0198 indicates well-calibrated predictions.

5) *Comparison and Analysis*: Table IV summarizes performance across all four benchmarks. STCA-ViT achieves state-of-the-art results on person-centric benchmarks (HMDB-51: 90.46%, UCF-101: 98.29%) where all modules operate effectively, while performance on large-scale diverse (Kinetics-400: 79.44%) and hand-centric (SSv2: 65.01%) datasets reflects module limitations in non-person-centric settings. Calibration is strong on HMDB-51 (ECE 0.0581) and SSv2 (0.0198), with higher ECE on UCF-101 (0.0888) and Kinetics-400 (0.1064) reflecting multi-class classification difficulty.

D. Ablation Study and Qualitative Analysis

1) *Module Contribution Analysis*: The contribution of each module in STCA-ViT is analyzed through an ablation study on HMDB-51. Table V presents the configurations and performance of variants A0–A11. All variants are measured after 15 epochs of training without TTA and are not directly comparable to the final performance (90.46% with TTA).

Backbone Baseline (A0). The VideoMAE ViT-Base backbone alone achieves 86.98%, confirming the strong foundation provided by Kinetics-400 pre-training representations.

Single Module Contribution (A1–A3). Each module independently improves performance. MGU provides the largest single improvement at $+0.90\%$ (A2, 87.88%), demonstrating the utility of Sobel-based explicit motion features, while STCA contributes $+0.64\%$ (A1, 87.62%) and PersonAttn contributes $+0.45\%$ (A3, 87.43%).

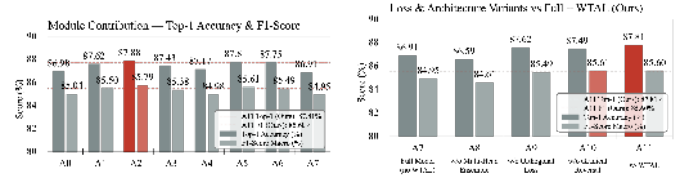


Fig. 8: Ablation results on HMDB-51. (Left) Module contribution (A0–A7). (Right) Loss and architecture variants (A7–A11).

Two-Module Combination (A4–A6). The STCA + PersonAttn (A5, 87.81%) and MGU + PersonAttn (A6, 87.75%) combinations show high performance. STCA + MGU (A4, 87.17%) performs lower than STCA alone (A1), which is attributed to optimization conflict (loss trade-off) during the short 15-epoch training, as both modules similarly capture short-term temporal variation information. With sufficient training epochs, the orthogonal loss and GRL progressively resolve this redundancy.

Full Module Integration (A7) and Variants (A8–A11). The full model (A7, 86.91%) is lower than some two-module combinations under the 15-epoch condition, but achieves 90.46% in the final evaluation with sufficient training and TTA. Removing the multi-head ensemble (A8, 86.59%) causes performance degradation, and removing the orthogonal loss (A9) lowers calibration quality by increasing ECE. Removing GRL (A10) weakens feature independence between modules, while adding the WTAL head (A11, 87.81%) improves upon A7 by $+0.90\%$, demonstrating that frame-level action supervision also contributes to classification performance.

2) *Attention Visualization*: Figure 9 shows MGU attention and person attention for the top 3 classes (100%) and bottom 3 classes (60–67%) on HMDB-51 test samples.

The MGU attention maps emphasize regions with large temporal variation—such as arm movements in *brush_hair*, object trajectories in *catch*, and hand movements in *clap*—while effectively suppressing backgrounds. The person attention module successfully localizes persons across diverse poses and scenes. Comparing the two streams reveals complementary characteristics: MGU focuses on motion dynamics while person attention focuses on spatial localization.

In failure cases, *fall_floor* (d) is misclassified as *kick* because the static posture immediately after a fall resembles the limb placement of kicking; *sit* (e) and *stand* (f) are misclassified as *drink* and *talk* respectively due to similar static postures. This reveals limitations for static actions with insufficient motion cues and suggests the need to integrate scene context and body pose information.

3) *Qualitative Analysis of WTAL Loss Terms*: Figure 10 compares WTAL loss terms before and after training on an HMDB-51 *throw* sample. After training, the MIL prediction shifts correctly to *throw*, background entropy is maintained at 95% of maximum (3.73 nats), and mean action score drops to 0.040, confirming sparsity. The pseudo-label loss dominates sparsity regularization due to the weight ratio ($\lambda_2=0.5$ vs. $\lambda_3=0.1$), and adaptive loss weighting remains a direction for future work.

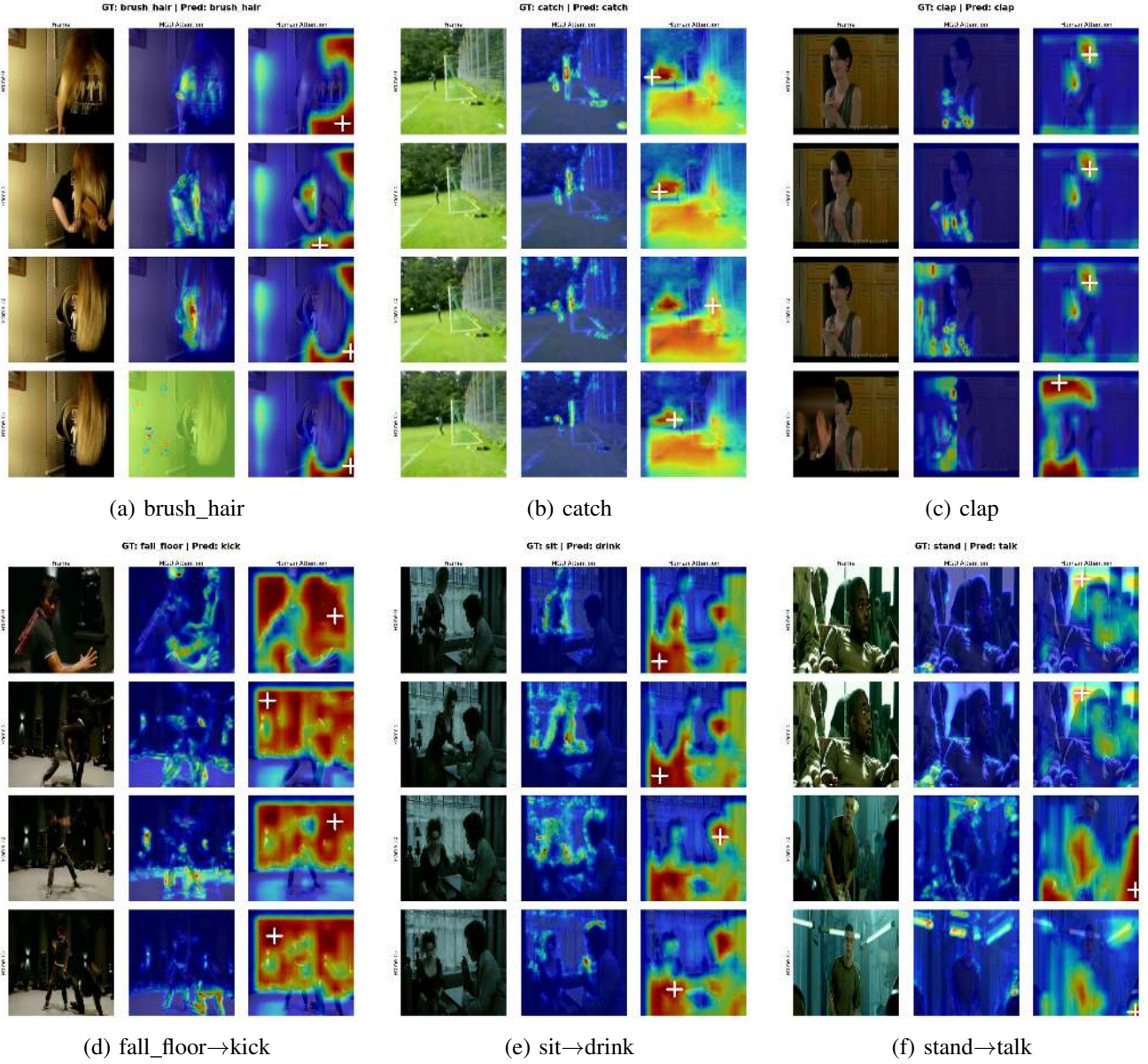


Fig. 9: Qualitative visualization of the attention mechanism of STCA-ViT on HMDB-51. Each example shows (left to right): original frame, MGU attention, and person attention. (a-c) Success cases (100% accuracy classes: brush_hair, catch, clap); (d-f) Failure cases (lowest accuracy classes: 60–67%): (d) *fall_floor* misclassified as *kick*, (e) *sit* misclassified as *drink*, (f) *stand* misclassified as *talk*.



Fig. 10: Before-and-after comparison of the WTAL head on an HMDB-51 *throw* class sample. (Left) Comparison of T-CAM, pseudo-target, and action score before training (Before, random initialization) and after training (After, trained checkpoint). (Right) Per-frame changes of 5 WTAL loss terms and a summary table. Orange = Before, Blue = After.

TABLE V: HMDB-51 ablation study. Top: configuration definitions for each variant (A0–A11). Bottom: performance by module combination based on the VideoMAE ViT-Base backbone (Top-1, Top-5, Macro F1, ECE). All results are measured after 15 epochs of training without TTA and are not directly comparable to the final STCA-ViT performance (90.46% with TTA).

ID	Modules	Description	Params (M)	Top-1 (↑)	Top-5 (↑)	F1 (↑)	ECE (↓)
<i>(a) Module Contribution</i>							
A0	Backbone only	ViT-Base + linear classifier	86.5	86.98	98.52	0.8504	0.1771
A1	+ STCA	Add STCA module	108.4	87.62	98.52	0.8550	0.1254
A2	+ MGU	Add MGU module	103.7	87.88	98.32	0.8579	0.1373
A3	+ Person	Add PersonAttn module	102.5	87.43	98.77	0.8538	0.1307
A4	+ STCA + MGU	No PersonAttn	110.2	87.17	98.52	0.8498	0.1369
A5	+ STCA + Person	No MGU	108.9	87.81	98.71	0.8561	0.1460
A6	+ MGU + Person	No STCA	104.2	87.75	98.39	0.8549	0.1347
A7	Full (no WTAL)	All modules, no WTAL head	110.7	86.91	98.65	0.8495	0.1310
<i>(b) Loss & Architecture Variants</i>							
A8	w/o Multi-Head	Single classifier head	109.4	86.59	98.58	0.8461	0.1148
A9	w/o Ortho Loss	Remove $\mathcal{L}_{ortho}$	110.7	87.62	98.71	0.8549	0.1388
A10	w/o GRL	Remove gradient reversal	110.7	87.49	98.77	0.8561	0.1346
A11	Full + WTAL	Add WTAL head	110.8	87.81	98.65	0.8560	0.1434

V. CONCLUSION

This paper proposes STCA-ViT, a video action recognition framework integrating six specialized modules: a VideoMAE ViT-Base backbone for high-level semantic feature extraction, a bidirectional spatio-temporal cross-attention module for joint spatial-temporal reasoning, a Sobel-based MGU for multi-scale motion analysis and boundary detection, a YOLO-based person attention module for person-centric spatial focus, a residual contribution fusion with gradient reversal for decorrelated feature integration, and a multi-head ensemble classifier with dynamic weights. The orthogonal loss enhances feature diversity across modules, and an optional WTAL head performs frame-level action detection without temporal annotations.

Experiments on four standard benchmarks confirm the effectiveness of STCA-ViT. It achieves 90.46% on HMDB-51, outperforming existing state-of-the-art models including VideoMAE V2 (ViT-g, 86.08%), and records 98.29% on UCF-101, 79.44% on Kinetics-400, and 65.01% on SSv2. These results are achieved with ~ 305 GFLOPs and ~ 110 M parameters, demonstrating an outstanding accuracy-efficiency balance compared to recent large-scale models requiring over 1 billion parameters and more than 500 GFLOPs. The primary limitation of STCA-ViT is its 16-frame design, which limits long-range temporal modeling as evidenced by lower performance on SSv2 (65.01%), and the dependence on full-body person detection, which reduces effectiveness in hand-centric datasets.

Future directions include evaluation of the WTAL head on untrimmed benchmarks (THUMOS14, ActivityNet), direction-aware and long-range temporal modeling for improved SSv2 performance, and multi-modal extension with edge deployment optimization.

ACKNOWLEDGMENT

This work was supported by the Ministry of Trade, Industry and Energy (MOTIE) and the Korea Evaluation Institute of Industrial Technology (KEIT), and by the National Research

Foundation of Korea (NRF) (Grant Nos. 2022R1A2C2005705 and RS-2024-00469138).

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